
Retrieval Relevance Is Not Verification Priority: Budgeted Provenance Checking in Inherited Agent Memory

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Abstract

1 A memory system that gives every record a provenance link is not thereby safe.
2 An agent inheriting hundreds of consolidated beliefs can follow only a few links
3 before it acts, so the safety-critical question is not whether provenance exists but
4 which links get followed. We study that allocation directly. In a controlled en-
5 vironment where exactly one of six inherited memories has lost a true negative
6 caveat, and the agent may pull at most k source records before committing, we
7 find allocation is governed by the agent’s own plan: across 1020 pre-registered
8 episodes the corrupted memory is verified in 236/236 episodes whose first-pass
9 intent it backs and 464/784 otherwise — complete separation, no counterexample.
10 Two new pre-registered experiments then establish that this allocation causes
11 later behaviour rather than merely correlating with it. Randomising which source
12 records are carried into a later decision, after the budget is spent and the situation
13 has shifted, moves the corrupted-direction rate from 139/150 to 3/150 (91 per-
14 centage points, every model). A third experiment separates what drives allocation
15 from what the memory means: with the agent’s plan held fixed, a memory carry-
16 ing an *irrelevant* hedge is checked far more often than the same memory with its
17 caveat silently deleted, so omission is stealthier than qualification. Finally, we test
18 the premise the threat model rests on by running a real consolidation chain, and
19 it fails: over six generations of re-summarisation, quantified negative outcomes
20 survive in every model, while normative prohibitions erode. The corruption that
21 matters is loss of the constraint, not loss of the fact. We argue persistent-memory
22 systems need a verification scheduler distinct from the retriever, and that the two
23 rank records differently.

24 1 Introduction

25 Long-horizon agents do not reason from evidence. They reason from memory: lessons written by
26 earlier sessions, compressed by consolidation passes, and inherited as durable organisational knowl-
27 edge. Provenance-aware memory designs answer this by keeping a link from every consolidated
28 record back to its source, so a suspicious belief can be checked.

29 That defence has a budget. An agent that inherits hundreds of beliefs and must act this cycle can
30 follow two or three links, not two hundred. Whether the memory system is safe therefore depends
31 on a decision the retrieval layer never makes: given many records that could have drifted and no
32 labels saying which did, *which records does the agent choose to audit?*

33 This is a ranking problem, and it is not the ranking a retriever computes. A retriever ranks by
34 relevance to the current query or plan. What an audit should rank by is the expected cost of the

record being wrong — which is often highest exactly where the agent is *not* currently looking, because a belief the agent is about to act on will be tested by acting, while a belief sitting quietly in memory will not.

We study this gap in a controlled setting. An agent inherits six length-balanced lessons, one of which has lost a true, materially negative caveat during consolidation. It may retrieve at most k source records before committing to an action. Because we removed the caveat, we know exactly which lookup recovers it, and every allocation is scored against that ground truth. The corrupted belief is deliberately *not* the one the situation makes salient: the symptoms point toward onboarding work, while the corrupted memory recommends promotional pricing.

Contributions.

1. **Problem formulation.** Budgeted selective verification over inherited memory as a reliability variable distinct from retrieval and from detection (§3).
2. **Measurement.** Verification allocation is governed by the agent’s current intent, with complete separation over 1020 pre-registered episodes (§4).
3. **Causal identification.** Two pre-registered experiments — a randomised-instrument replay and a randomised carry-forward — show that what the budget recovered *causes* the later decision, with the confounds of the observational design removed by construction (§5).
4. **Mechanism.** With intent held fixed, allocation responds to unresolved qualification, and a silently deleted caveat attracts *less* scrutiny than an irrelevant hedge (§6).
5. **A negative result that reframes the threat model.** Real consolidation does not produce the corruption this literature assumes. Over six generations, quantified negatives survive; prohibitions erode (§7).

Study 1 (§4) re-analyses episode files released with an earlier preprint by the same authors; the citation is omitted here for anonymity and will be restored in the camera-ready. Studies 2–4 are new, pre-registered before any model call, and reported with their failures.

2 Related work

Memory integrity. Memory-poisoning work studies adversarial insertion into agent memory [Dash et al., 2026, Xie et al., 2026] and defences that trust sources rather than content [Singh, 2026]. Our corruption is not adversarial: nothing false is inserted, and the drifted summary is what a benign compression pass produces. The threat model is operational, extending fault injection [Tan et al., 2026] from transport faults to the semantic layer.

Budgeted verification. A growing line allocates limited verification compute across candidate *outputs* or reasoning steps [Yuan et al., 2026a, Fang et al., 2026, Yang, 2026], and benchmarks ask whether agents manage tool budgets at all [Wang and Xu, 2026, Lin et al., 2026]. We allocate over *inherited beliefs* and measure the direction of the allocation rather than its total.

Provenance and self-auditing. Execution-provenance surveys catalogue how evidence and actions connect in agent traces [Wang et al., 2026]; self-auditing repairs unsupported intermediate beliefs [Yuan et al., 2026b], and provenance-sensitivity audits ask whether action selection responds to source information at all [Liao, 2026]. These assume the audit target is given, or audit everything. We study the upstream selection problem.

Positive test strategies. That an agent checks the evidence behind its favoured hypothesis rather than evidence that could overturn it rhymes with human hypothesis-testing research [Wason, 1960, Klayman and Ha, 1987]. We claim no shared mechanism; we claim the allocation signature is measurable and steerable in agents.

3 Setup

A growth agent for a synthetic B2B SaaS business inherits six durable memories, each a one-to-two-sentence lesson with a consolidation date and an archived source. Bodies are length-balanced (123–

Table 1: Firth-penalised logistic model of whether the corrupted memory’s source was requested, over all 1020 Study-1 episodes. Robustness and synthesis; the pre-registered contrasts remain the confirmatory analysis.

term	odds ratio	95% CI
intent aligned	422	[26, 6860]
caveat deleted	4.44	[2.73, 7.20]
triage invariant	13.2	[7.60, 22.97]
budget (+1 credit)	13.2	[6.98, 24.81]
target position	0.99	[0.89, 1.11]
corruption moved onto the intended path	3.90	[1.13, 13.47]

132 characters) and identically formatted; presentation order is shuffled per episode with a seeded PRNG. The situation presents three concurrent symptoms so that four of the six memories bear on something and no action trivially dominates. The agent may retrieve at most k source records by id, then commits to one action. Six models from two providers are used throughout, at their default or medium reasoning setting, with byte-identical prompts and structured output enforced.

In the *clean* arm the target memory carries its true caveat; in the *drifted* arm the caveat is absent and the remaining sentence is still true. A *triage* arm adds one system-prompt sentence: *when verification capacity is limited, prioritize inherited beliefs that are both highly consequential to the current decision and insufficiently supported for the objective being optimized*. It names no memory, metric, or direction.

4 Study 1: allocation follows intent

Across all 1020 episodes — every condition, budget, target and model — the corrupted memory is verified in **236/236** episodes whose first-pass intended action it backs, against 464/784 (59%) otherwise. The relation is completely separated: the corpus contains no episode in which the agent planned to act on the corrupted belief and did not check it.

Table 1 fits one model to all 1020 episodes. Because intent-alignment separates the outcome perfectly the ordinary MLE diverges, so we use Firth’s penalised likelihood; the separation is the finding, not a nuisance. Target position carries an odds ratio of 0.99, confirming that randomising presentation order removed it as a confound.

A uniform policy at $k=2$ over six memories reaches the target 33% of the time. Models are better than that on average and far better when the target is in path — and worse than uniform in the models at the bottom of the spread. The aggregate hides the structure: allocation is not weakly risk-sensitive, it is strongly plan-sensitive.

5 Study 2: the allocation causes the later decision

Study 1 shows where the budget goes. It does not show that this matters. The natural test — let the situation shift after the budget is spent, then see whether episodes that recovered the caveat behave differently — is observational: verification is chosen by the model, so the two groups differ in whatever made the model choose.

An earlier two-phase experiment ran exactly that comparison and reported a large split. Auditing its implementation we found two paths that the design left open and the reporting did not separate. First, the triage arm’s system prompt was constructed once and reused for the later call, so the randomised instruction was still in context when the second decision was made and could act on it directly. Second, the phase-1 conversation was carried forward, so the model saw its own earlier reasoning — which in the drifted arm frequently argued explicitly against the corrupted direction. “Verified” and “unverified” episodes therefore differed in what the model had already committed to in writing, not only in what evidence it held.

We close both by construction, pre-registering two experiments before any call.

Table 2: Study 2b. Corrupted-direction rate when the provenance carried into the later decision is randomised rather than chosen. Both arms carry two source records; only whether one of them is the corrupted memory’s differs.

	corrupted memory’s source carried	not carried
Opus 5	0/25	14/25
Sonnet 5	2/25	25/25
Haiku 4.5	0/25	25/25
Sol	0/25	25/25
Terra	1/25	25/25
Luna	0/25	25/25
pooled	3/150 (2%)	139/150 (93%)

5.1 A randomised-instrument replay

For each of the 450 episodes of the earlier experiment we replay *only* the second decision, as a fresh session: the triage sentence is absent from every arm, no conversation history is carried, the memory block and its per-episode order are reproduced, and the only thing that differs is a RECOVERED block holding source records for exactly the memories that episode’s budget was actually spent on. The situation text is byte-identical across arms.

Triage assignment was randomised upstream and moves recovery hard. In the replay it cannot reach the outcome by any route except through what was carried forward, because the session is otherwise identical across arms. This is a randomised encouragement design in which the exclusion restriction is true by construction rather than assumed.

It replicates. The corrupted-direction rate is 13/150 (9%) under triage against 55/150 (37%) without it (CMH stratified by model, $p = 2.2 \times 10^{-9}$). And the exclusion restriction can be checked rather than argued: matched on what phase 1 recovered, the two arms are indistinguishable in both strata — 4/94 versus 3/140 among episodes that recovered the caveat ($p = 0.764$), and 51/56 versus 10/10 among those that did not ($p = 0.61$). With the instruction removed from the room, arms holding the same evidence behave the same way.

The Wald estimate of the effect of recovering the corrupted memory’s source is -0.91 (95% CI $[-1.34, -0.62]$ by stratified bootstrap over models): close to complete compliance, an agent that recovers the caveat essentially does not walk into the corrupted direction.

The original design was conservative, not inflated. Removing the carried-forward conversation *raises* the corrupted-direction rate among episodes that never recovered the caveat, from 79% in the original to 91% here. The model’s own earlier prose had been suppressing the choice, and the original design credited that suppression to provenance. The confound worked against the reported effect.

5.2 Randomising the provenance itself

The replay still lets the model choose what it recovered. So we also assign it. In 300 fresh single-call episodes the memory block is the drifted lineage in both arms, and two source records are carried forward: in CARRY-TARGET one of them is the corrupted memory’s, in CARRY-OTHER neither is. Both arms carry exactly two records; only their identity differs, and it is assigned by seeded PRNG.

Table 2 gives the result: 3/150 against 139/150, a difference of 91 percentage points (CMH $\chi^2 = 246$, $p < 10^{-54}$), in the same direction in 6 of six models, with five of six choosing the corrupted direction in every single episode where its provenance was absent.

Two independent identification strategies — an instrument in 2a, full randomisation in 2b — put the effect at the same place, near -91 percentage points. What the budget happened to recover is not a marker of some other difference between episodes. It is the thing that decides.

What this does and does not establish. It establishes that *holding* the recovered caveat changes the later decision. It does not establish that models allocate budget *in order to* secure future evidence;

Table 3: Study 3. Verification of the corrupted memory, first pass, budget two, by what its body carries. Bodies are matched on length and clause structure across the three appended arms; only the content differs.

	drifted (no qualifier)	clean-positive (positive)	clean-neutral (weakens confidence)	clean-negative (true caveat)
verified	83/126 (66%)	118/126 (94%)	98/126 (78%)	28/103 (27%)

Study 1 says the opposite, that they allocate toward the present plan. The causal chain we have closed is allocation → what is in context later → behaviour, not any claim about foresight.

6 Study 3: what allocation actually responds to

Study 1 contrasts a drifted memory against a clean one carrying its caveat, and reads the difference as detection of the omission. Auditing the construction we found that contrast varies four things at once. Appending the caveat makes the target body 221 characters against 123–132 for every other memory — $1.73\times$ — and makes it the only memory in the set carrying a *However* clause. Length, surface qualification, and uniqueness move with correctness.

So we separate them. Four arms run concurrently, differing only in what is appended to the corrupted memory’s body, matched to the true caveat on character length and on clause structure:

- **drifted** — nothing appended (129 characters).
- **clean-positive** — a positive elaboration, no hedge (220).
- **clean-neutral** — a measurement-hygiene note that weakens confidence in the finding without saying the action is risky (221).
- **clean-negative** — the true caveat (221).

Table 3 orders the arms in a way the salience account does not predict. The true caveat does not raise verification; it *collapses* it (27%). A matched-length qualifier that does not answer the question raises it (78% and 94%). And the silently drifted memory sits below both (66%).

The mechanism is visible in what the budget goes to instead. Holding one model fixed, the first-pass intent was `simplify_onboarding` in every episode of all four arms — the plan does not move — yet the allocation does. With the true caveat present the two credits go to the memories backing that plan and the corrupted memory is never requested. With a qualifier that raises a question it does not settle, a credit is spent on it even though the agent does not intend to act on it.

Two things follow.

The published contrast holds, but not for the stated reason. Verification really is higher when the caveat is absent than when it is present (1.1×10^{-13} and $< 10^{-22}$ for the matched arms against the true caveat). But this is not the agent noticing that something was removed. It is the agent not needing to look at a belief whose constraint is already written down. Detection of the omission is not required to produce the effect, and Study 3 gives no evidence that it occurs.

Silence is stealthier than qualification. A memory that hedges about something irrelevant draws more scrutiny than one whose material caveat was deleted (0.018). Consolidation that drops a clause therefore does not merely remove information — it removes the surface cue that would have made the record look worth checking. The record becomes cleaner-looking and correspondingly less audited, which is the worst combination for a store that future sessions will inherit.

7 Study 4: does benign consolidation actually produce this?

Studies 1–3 install the corruption by hand, which buys clean ground truth. It also assumes something the literature repeats and, as far as we can tell, has not measured: that summarisation drops caveats. We tested it, expecting confirmation, and did not get it.

194 We ran a real consolidation chain over the same six source records, with no instruction to preserve or
195 drop anything and no mention of caveats, qualifiers, or what matters: an analyst writes session notes;
196 the notes are consolidated into durable lessons; the durable memory is then re-summarised four
197 times as it ages and later cycles compete for context, with targets stepping 130 → 110 → 90 → 70
198 characters. 60 chains, six generations each, across six models.

199 Survival is scored by a model that sees the original source record and all six generations at once and
200 reports, for each, whether the source’s specific constraint is still stated. Two of the six records carry
201 no qualifier at all and act as a false-positive check: the scorer identified them as unqualified in every
202 case and reported a constraint in 0/60 judgments where none exists.

203 **The premise fails for negative outcomes.** A quantified negative result is close to the last thing
204 compression discards. At the tightest compression — lessons cut to around 70 characters — the
205 retention harm is still stated in 73% of chains. Several models reach that budget with sentences of
206 the form “40% SMB discount: signups +31%, retention −12%”, having dropped everything else.
207 The number is load-bearing enough to be kept.

208 **What does erode is scope, and normative force.** Scope restrictions go roughly twice as fast: 44%
209 survive at the same compression against 73% for the negative outcome. And within the corrupted
210 memory itself, separating the finding from the instruction shows the split directly: at the consoli-
211 dation stage 60/60 chains state the negative outcome and 37/60 still state a prohibition; after four
212 re-summarisations, 60/60 still state the outcome and only 22/60 state a prohibition. What consoli-
213 dation removes is not the fact that discounting hurt retention. It is the sentence saying not to do it
214 anyway.

215 **Consequences for the threat model, including ours.** The corruption Studies 1–3 use — deleting
216 an entire negative caveat — is not what this process produces. A benign-compression threat model
217 should be built on the operators compression actually applies: loss of scope, and loss of normative
218 constraint while the supporting number survives. Those are more dangerous than they sound, be-
219 cause a lesson that retains its numbers looks well-evidenced. Our own Study 3 says such a record
220 draws less verification, not more.

221 We report this as a limitation on Studies 1–3 rather than repairing it here. The allocation results stand:
222 they concern which beliefs an agent checks given a corrupted one, and hold for any corruption with
223 the same decision geometry. What does not stand is the claim that this particular corruption arises
224 on its own.

225 **A pre-registered rule that failed.** We registered that two scorers — a keyword rule and the model
226 scorer — must agree. That rule is mis-specified and we report it rather than its output. Disagree-
227 ments occur precisely where one scorer sees a lost qualifier, so conditioning on agreement censors
228 informatively and returns 100% survival at every generation by construction. The keyword rule also
229 turned out to have very low specificity on scope restrictions, firing on any mention of the segment
230 term, so it functions only as an upper bound. We report the model scorer, validated against the un-
231 qualified controls, as primary, and state the bound rather than presenting the agreeing subset as an
232 estimate.

233 8 Discussion: memory systems need a verification scheduler

234 The results describe a control problem that current memory architectures do not expose. A retrieval
235 layer answers *which records are relevant to what the agent is doing*. Nothing answers *which records*
236 *are worth spending scarce provenance checks on*. In our environment those two rankings are close
237 to opposites: the belief most costly if wrong is the one outside the current plan, and that is exactly
238 the one the budget does not reach.

239 Making the second ranking explicit needs metadata the first does not: how many consolidation
240 generations a record has passed through, when its provenance was last actually followed, how broad
241 its scope claim is relative to the evidence behind it, and which future contexts could make it load-
242 bearing. Study 4 gives a concrete signal for the third: a record that retains a number while losing
243 its scope or its prohibition has drifted in the direction that matters, and the drift is detectable by

244 comparing generations, which a consolidation pipeline is in a position to do and an agent reading
245 the final summary is not.

246 We are wary of the term *verification debt*, but the accounting it suggests is the useful part. A durable
247 belief that survives many consolidations without its provenance ever being followed accumulates
248 risk that no single session sees, because each session rationally spends its budget on its own plan.
249 Study 1 shows that behaviour is not a defect of any one model — it is what every model we tested
250 does, and it is locally correct. The fix is therefore unlikely to be a better-behaved agent. It is a
251 scheduler outside the agent that can spend verification against the store’s accumulated exposure
252 rather than against today’s decision.

253 **Steering works, partially.** One system-prompt sentence naming no memory and no direction
254 moves allocation substantially (Table 1, odds ratio 13.2), and Study 2a shows that this reaches be-
255 haviour one decision later through what it caused the agent to recover. But it does not equalise
256 models, and in our replay the model with the weakest allocation still walks into the corrupted di-
257 rection a quarter of the time under the instruction. Prompt-level triage is worth writing and is not a
258 substitute for scheduling.

259 9 Limitations

260 **One scenario.** Every episode instantiates the same synthetic growth-agent situation with six mem-
261 ories and five actions. We deliberately did not add a second domain: built in the time available it
262 would have been a paraphrase of the first, and generality claimed from two structurally identical
263 scenarios is worse than generality not claimed. Domain replication and larger memory pools are the
264 obvious next step and we have not taken it.

265 **Small memory pool, perfect archive.** Six records, explicit ids, and a provenance store that always
266 answers correctly and instantly. Real stores are larger, slower, sometimes missing, and sometimes
267 themselves stale. We expect all of these to sharpen the allocation problem, but we have not shown
268 it.

269 **Study 1 is a re-analysis.** Its episodes were collected earlier under their own pre-registration. The
270 confounds we identified in its clean arm (Study 3) and in its two-phase follow-up (Study 2) are why
271 Studies 2–4 exist; where Study 1’s reported interpretation and our new evidence conflict, we have
272 said so in the section concerned rather than restating the earlier reading.

273 **Causal scope.** Studies 2a and 2b identify the effect of *holding* recovered provenance on the next
274 decision. Neither identifies why models allocate as they do, and nothing here tests whether allocation
275 is stable across tasks or sessions — so we avoid calling it a trait.

276 **Behaviour, not harm.** We measure which direction the agent chooses and at what scale. Under
277 a strict pre-registered harm rule requiring an unhedged, unverified, corrupted-evidence rollout, no
278 episode in this work or its predecessor qualifies: models hedge verbally while still choosing the
279 direction. That is itself a warning about self-reported uncertainty as a safety metric, and it means
280 our outcome is exposure, not damage.

281 10 Conclusion

282 Provenance links do not make an inherited memory store safe; following them does, and following
283 is budgeted. We find that agents spend that budget on the beliefs behind the plan they already
284 have — with no counterexample in 1020 episodes — and that what the budget happens to recover
285 causally determines what they do when the situation later shifts, by roughly ninety percentage points
286 under two independent randomised designs. We also find that a silently deleted caveat attracts less
287 scrutiny than an irrelevant hedge, and that real consolidation does not delete caveats so much as
288 strip their scope and their normative force while keeping the numbers that make them look well-
289 evidenced. Retrieval relevance and verification priority are different rankings. Memory systems
290 currently compute only the first.

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